PROJECT REPORT (AI AND ML)

NAME-ARYAN GUPTA

REGISTRATION NUMBER-25BAI11516

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1. Introduction

This project, titled *Smart Student Assistant*, is an attempt to apply Artificial Intelligence and Machine Learning concepts to solve a real-world academic problem. The system is designed to help students understand their academic performance and provide suggestions for improvement.

The project integrates Machine Learning for prediction and Prolog for logical reasoning, demonstrating how different AI techniques can be combined to build an intelligent system.

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2. Problem Identification

During college life, it is common to observe that many students are unsure about how their study habits affect their academic performance. Students often ask questions like:

- "Am I studying enough?"
- "Why are my marks low?"
- "What should I improve?"

However, there is no simple tool that gives both:

- A prediction of performance
- Clear suggestions for improvement

This gap led to the idea of building a system that can assist students using data and logic.

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3. Why This Problem Matters

This problem is important because:

- Academic performance directly impacts student confidence and future opportunities
- Students often lack personalized guidance
- Early feedback can help students improve before exams

By solving this problem, the project aims to provide students with a simple tool to make better academic decisions.

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4. Approach to the Solution

To solve the problem, a hybrid AI approach was used:

Step 1: Machine Learning

A Linear Regression model is used to predict marks based on:

- Study hours
- Attendance

This provides a numerical estimate of performance.

Step 2: Rule-Based Reasoning (Prolog)

A Prolog-based system is used to analyze conditions such as:

- Low attendance
- Low marks

Based on these conditions, it generates suggestions like:

- "Improve attendance"
- "Study more regularly"

Step 3: Integration

Python is used to integrate both components into a single system that takes input and produces output.

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5. Key Design Decisions

Several important decisions were made during development:

- **Choice of Linear Regression:**
  Selected because it is simple, easy to implement, and suitable for beginners.
- **Use of Prolog:**
  Included to demonstrate knowledge representation and logical reasoning from the syllabus.
- **Command-Line Interface:**
  Chosen to keep the project simple and focused on core logic rather than UI design.
- **Small Dataset:**
  Used for simplicity and quick model training.

These decisions ensured that the project remained manageable while still covering important AI concepts.

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6. Implementation Overview

The system is divided into three main parts:

1. Machine Learning Module

- Trains a model using student data
- Predicts marks

2. Prolog Module

- Contains rules for decision-making
- Generates advice

3. Integration Module

- Takes user input
- Calls ML model
- Calls Prolog system
- Displays results

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7. Challenges Faced

Several challenges were encountered during the project:

- Difficulty in understanding Prolog syntax
- Issues in integrating Prolog with Python
- Setting up the SWI-Prolog environment
- Debugging errors in code

These challenges helped in gaining practical experience and problem-solving skills.

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8. Results

The system successfully:

- Predicts marks based on input values
- Provides relevant suggestions
- Demonstrates the working of AI techniques

Example:

Input:

- Study Hours: 5
- Attendance: 70

Output:

- Predicted Marks: 64

- Advice: Improve attendance

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9. What I Learned

This project helped in understanding:

- Basics of Machine Learning
- How Linear Regression works
- Prolog programming and logic rules
- Integration of different technologies
- Real-world application of AI concepts

It also improved problem-solving and debugging skills.

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10. Conclusion

The Smart Student Assistant project demonstrates how Artificial Intelligence and Machine Learning can be applied to solve real-world academic problems.

It not only predicts student performance but also provides meaningful suggestions, making it a useful and practical tool. The project also reflects the importance of combining different AI techniques to build intelligent systems.